

General Parametric Equations: The Cycloid

General Idea

Generally, parametric equations describe arbitrary motion in the plane or in space.

Cycloid

A cycloid is obtained by a wheel of radius a rolling along a straight line (the x -axis).

Let P be a point on the rim of the wheel. Assume that P starts at the origin.

Problem

Find the position $(x(t), y(t))$ of the point P as a function of the angle θ through which the wheel has rotated.

Derivation

We express the position vector of P as:

$$\vec{OP} = \vec{OA} + \vec{AB} + \vec{BP}$$

The horizontal displacement is equal to the arc length rolled:

$$OA = a\theta$$

Thus, the parametric equations of the cycloid are:

$$x(\theta) = a(\theta - \sin \theta)$$

$$y(\theta) = a(1 - \cos \theta)$$

Vector Components

We decompose the vectors:

$$\vec{AB} = \langle 0, a \rangle$$

The vector $\vec{BP}$ has magnitude a and makes an angle θ with the vertical, hence:

$$\vec{BP} = \langle -a \sin \theta, -a \cos \theta \rangle$$

Thus,

$$\vec{OP} = \langle a\theta - a \sin \theta, a - a \cos \theta \rangle$$

So the parametric equations are:

$$x(\theta) = a\theta - a \sin \theta$$

$$y(\theta) = a - a \cos \theta$$

Behavior Near the Bottom

Take the radius as the unit length, i.e., $a = 1$.

Then:

$$x(\theta) = \theta - \sin \theta, \quad y(\theta) = 1 - \cos \theta$$

Taylor Approximation

For small θ , we use the Taylor expansions:

$$f(t) \approx f(0) + tf'(0) + \frac{t^2}{2!}f''(0) + \frac{t^3}{3!}f'''(0)$$

In particular:

$$\sin \theta \approx \theta - \frac{\theta^3}{6}$$

$$\cos \theta \approx 1 - \frac{\theta^2}{2}$$

Approximate Cycloid Near $\theta = 0$

Substitute into the parametric equations:

$$x(\theta) \approx \theta - \left(\theta - \frac{\theta^3}{6} \right) = \frac{\theta^3}{6}$$

$$y(\theta) \approx 1 - \left(1 - \frac{\theta^2}{2} \right) = \frac{\theta^2}{2}$$

Vector Decomposition

We have:

$$\vec{AB} = \langle 0, a \rangle$$

The vector $\vec{BP}$ has magnitude a and makes an angle θ with the vertical. Hence,

$$\vec{BP} = \langle -a \sin \theta, -a \cos \theta \rangle$$

Therefore,

$$\vec{OP} = \vec{OA} + \vec{AB} + \vec{BP} = \langle a\theta - a \sin \theta, a - a \cos \theta \rangle$$

So,

$$x(\theta) = a\theta - a \sin \theta, \quad y(\theta) = a - a \cos \theta$$

Question

What happens near the bottom of the cycloid?

Answer

Take the unit length equal to the radius, i.e. $a = 1$:

$$x(\theta) = \theta - \sin \theta, \quad y(\theta) = 1 - \cos \theta$$

Taylor Approximation

For small θ , Taylor expansion gives:

$$f(t) \approx f(0) + tf'(0) + \frac{t^2}{2!}f''(0) + \frac{t^3}{3!}f'''(0)$$

In particular,

$$\sin \theta \approx \theta - \frac{\theta^3}{6}, \quad \cos \theta \approx 1 - \frac{\theta^2}{2}$$

Approximation Near $\theta = 0$

Substituting:

$$x(\theta) \approx \theta - \left(\theta - \frac{\theta^3}{6} \right) = \frac{\theta^3}{6}$$

$$y(\theta) \approx 1 - \left(1 - \frac{\theta^2}{2} \right) = \frac{\theta^2}{2}$$

The Position Vector

In order to use vector techniques, we define the position vector:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} = \langle x(t), y(t), z(t) \rangle$$

This is just the vector from the origin to the moving point. As the point moves, so does the position vector (see the figure with example 1).

Example 1

Thomas Pynchon fires a rocket from the origin. Its initial x -velocity is v_{0x} and its initial y -velocity is v_{0y} .

$$x(t) = v_{0x}t, \quad y(t) = -\frac{1}{2}gt^2 + v_{0y}t$$

At time t , the rocket is at point

$$P = (x(t), y(t))$$

The position vector can be written in many different ways:

$$\mathbf{r}(t) = \overrightarrow{OP} = x(t)\mathbf{i} + y(t)\mathbf{j} = \langle x, y \rangle$$

Circles and Ellipses

Consider the parametric curve in the plane:

$$x(t) = a \cos t, \quad y(t) = a \sin t$$

Easily, we have the relation:

$$x^2 + y^2 = a^2 \cos^2 t + a^2 \sin^2 t = a^2$$

Therefore, the trajectory is a circle of radius a centered at 0.

We will call $x(t) = a \cos t$, $y(t) = a \sin t$ the parametric form of the curve, and

$$x^2 + y^2 = a^2$$

the symmetric form.

A Different Parametrization

Note a different parametrization:

$$x(t) = a \cos(3t), \quad y(t) = a \sin(3t)$$

This results in the same path, i.e., the circle

$$x^2 + y^2 = a^2$$

but the two trajectories differ by how fast they travel around the circle.

Ellipse

The circle generalizes easily to an ellipse with parametric form:

$$x(t) = a \cos t, \quad y(t) = b \sin t$$

and symmetric form:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$